

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

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In [ ]: df = pd.read_csv("train.csv")
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In [ ]: df.shape
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In [ ]: df.head()
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In [ ]: df.info()
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In [ ]: df.columns
```

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In [ ]: df.describe()
```

```
In [ ]: df.isna().sum()
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```
In [ ]: df["Age"] = df["Age"].fillna(df["Age"].mean())
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```
In [ ]: df.isna().sum()
```

```
In [ ]: df["Name"]
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In [ ]: df["Sex"].value_counts()
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In [ ]: df["Ticket"].value_counts()
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In [ ]: df["Cabin"].value_counts()
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In [ ]: df["Embarked"].value_counts()
```

```
In [ ]: def fun1(value):
    if (value == "male"):
        return 1
    else:
        return 0
```

```
In [ ]: def fun2(value):
    if (value == 'S'):
        return 0
    elif (value == 'C'):
        return 1
    elif (value == 'Q'):
        return 2
    else:
        return 0
```

```
In [ ]: df["Sex"] = df["Sex"].apply(fun1)
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In [ ]: df["Embarked"] = df["Embarked"].apply(fun2)
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In [ ]: df.isna().sum()
```

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In [ ]: df = df.drop("Cabin", axis=1)
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In [ ]: df.shape
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In [ ]: df.shape
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```
In [ ]: #Set up the figure and axes
fig, axes = plt.subplots(1, 3, figsize=(18, 6))
# Age Distribution
sns.histplot(data=df, x='Age', kde=True, ax=axes[0])
axes[0].set_title('Age Distribution')
# SibSp Distribution
sns.histplot(data=df, x='SibSp', kde=True, ax=axes[1])
axes[1].set_title('SibSp Distribution')
# Parch Distribution
sns.histplot(data=df, x='Parch', kde=True, ax=axes[2])
axes[2].set_title('Parch Distribution')
#plt.tight_layout()
#plt.show()
# Fare Distribution
plt.figure(figsize=(10, 6))
sns.histplot(data=df, x='Fare', kde=True)
plt.title('Fare Distribution')
plt.show()
```

```
In [ ]: df.info()
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In [ ]: sns.countplot(df, x="Survived")
plt.show()
```

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In [ ]: sns.countplot(df, x="Pclass", hue="Survived", palette="Accent")
plt.show()
```

```
In [ ]: sns.countplot(df, x="Embarked", hue="Survived")
plt.show()
```

```
In [ ]: sns.histplot(df["Fare"])
plt.show()
```

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In [ ]:
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